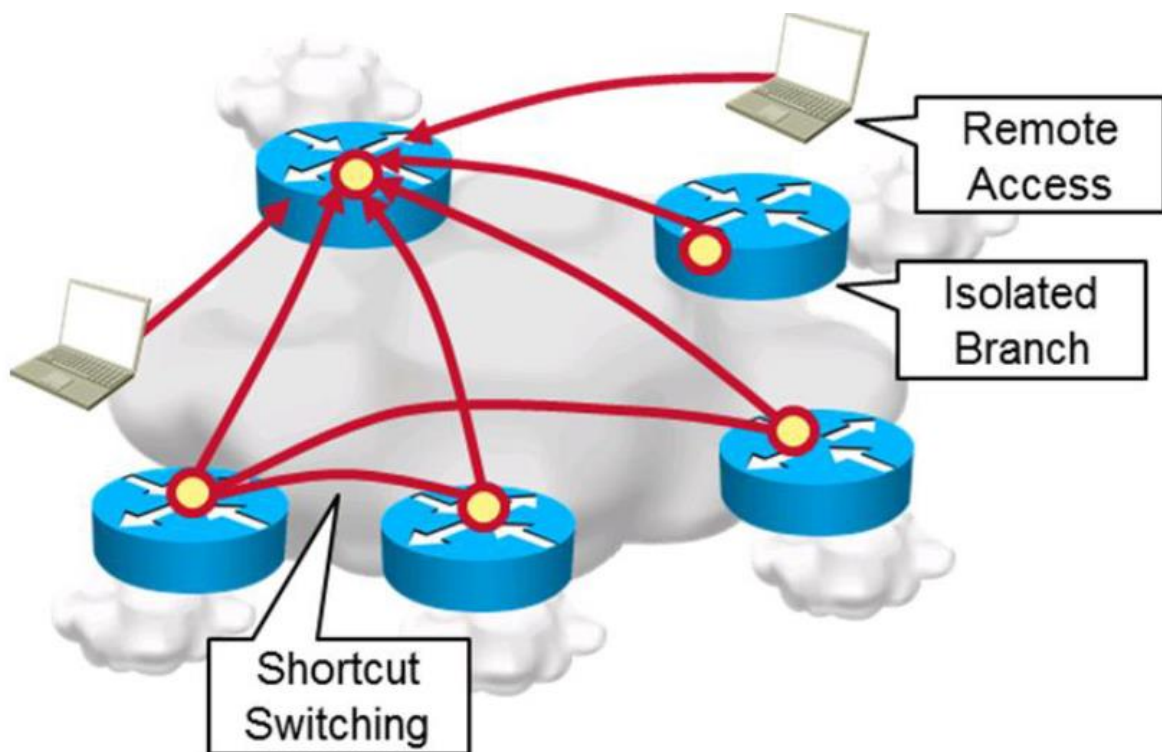
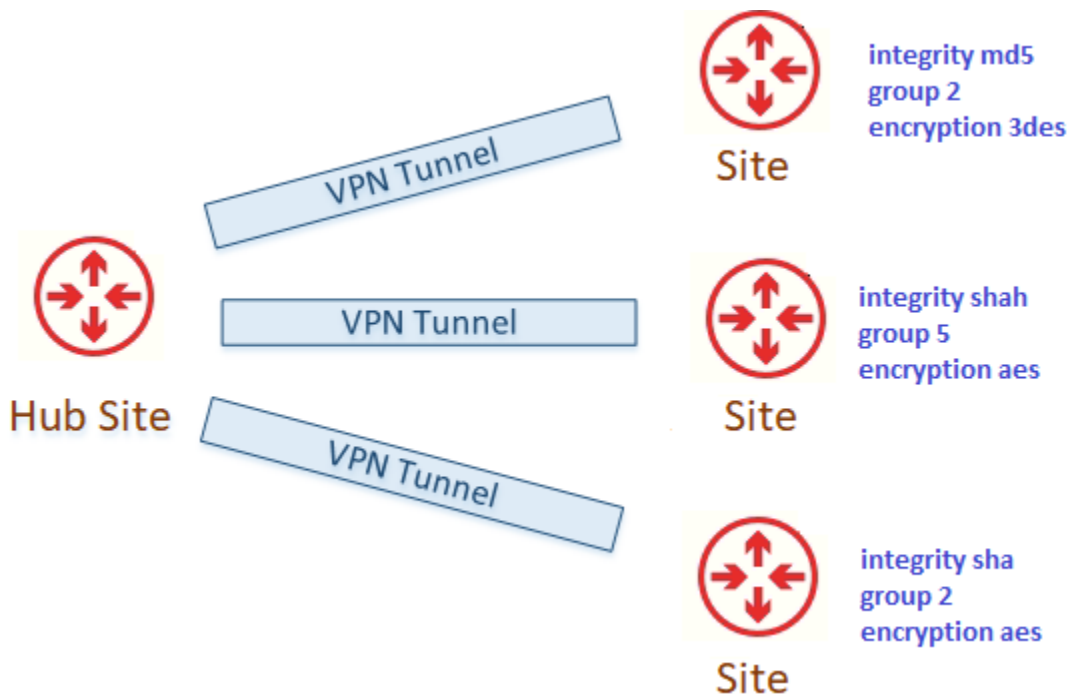


FlexVPN:

- o FlexVPN is a framework to configure IPsec VPNs on Cisco IOS devices only.
- o FlexVPN was created to simplify the deployment of VPN solutions of all type.
- o Such as Remote Access, Site-to-Site and Dynamic Multipoint VPN topologies.
- o FlexVPN uses common configuration template for all types of VPN topologies.
- o FlexVPN is based on Internet Key Exchange (IKEv2) and does not support IKEv1.
- o FlexVPN is the Cisco's way of configuring Internet Key Exchange Version 2 (IKEv2).
- o Most of configuration commands begin with crypto Internet Key Exchange Version 2.
- o IKEv2 come with "Smart Defaults" representing Cisco's view of best practice design.
- o Need single interface template on Hub to allow all types of incoming VPN connections.
- o FlexVPN, unofficially called the Dynamic Multipoint Virtual Private Network Phase 4.
- o Flexible VPN common umbrella term for all IKEv2 IPsec VPNs deployed on IOS routers.
- o Flexible VPN (FlexVPN) is a latest way of configuring IP Security (IPsec) VPN with IKEv2.
- o When use IKEv2 & deploy Site-to-Site Virtual Private Network, its call Flex Site-Site VPN.
- o When use IKEv2 & deploy Remote Access VPN, we call it Flex Remote Access VPN.
- o IKEv2 simplifies the message exchange and provides support for Voice over IP traffic.
- o Flex VPNs offer unified and modular framework for all type of VPN implementation.
- o Implement FlexVPN Site to Site using Crypto Maps, static VTI, DVTI & GRE IPsec tunnels.
- o FlexVPN is new framework to configure IPsec VPN with IKE version 2 on IOS platforms .





VTI (Virtual Tunnel Interface):

- o VTI stands for Virtual Tunnel Interface is a full-featured routable interfaces.
- o Many options applied to physical interfaces can be applied to virtual tunnel interface.
- o IPsec Virtual Tunnel Interface (VTIs) support multicast traffic such as voice and video.
- o IPsec Virtual Tunnel Interface (VTIs) require fewer SAs to support all types of traffic.
- o Like GRE VTIs support all types of IP routing protocols, provides scalability & redundancy.
- o Dynamic VTI tunnels should be for the hub in large Hub-and-Spoke implementations.
- o Static VTI should be for the Spoke only in large Hub-and-Spoke implementations.
- o SVTI configurations can be used for Site-to-Site connectivity tunnel always-on access.
- o Enable dynamic routing protocols on tunnel interface without extra 24 bytes like GRE.
- o Static VTI configuration reducing the bandwidth for sending the encrypted data.
- o DVTI technology replaces dynamic crypto maps & the Dynamic Hub-and-Spoke method.
- o DVTI technology replaces dynamic crypto maps & Hub-&Spoke for establishing tunnels.
- o DVTI requires smaller header 4 bytes compared to DMVPN which is additional 28 bytes.
- o Static Virtual Tunnel Interface configurations can be used for site-to-site connectivity.
- o Static Virtual Tunnel Interface (SVTI) provides always-on access between two sites.
- o DVTIs can provide highly secure and scalable connectivity for remote-access VPNs.
- o The DVTI (Dynamic VTIs) can be used for both the server and remote configuration.
- o The tunnels provide an on-demand separate virtual access interface for each VPN session.
- o Configuration of virtual access interfaces is cloned from virtual template configuration.
- o Which includes the IPsec configuration and any Cisco IOS software feature configured.
- o Dynamic VTIs function like any other real interface so that you can apply QoS, firewall etc.

Smart Defaults:

- o The FlexVPN configuration can be minimized using the IKEv2 Smart Defaults.
- o These specify default values for all components except IKEv2 Profile & Keyring.
- o The Smart defaults configuration can be modified as per your requirements.
- o The Smart defaults can be displayed if use the command “show running-config all”.
- o The default configuration can be disabled by using “no” before the command.
- o The default mode for the default transform set is “Transport”, mode.
- o Whereas all other transform sets the default mode is “Tunnel” mode.

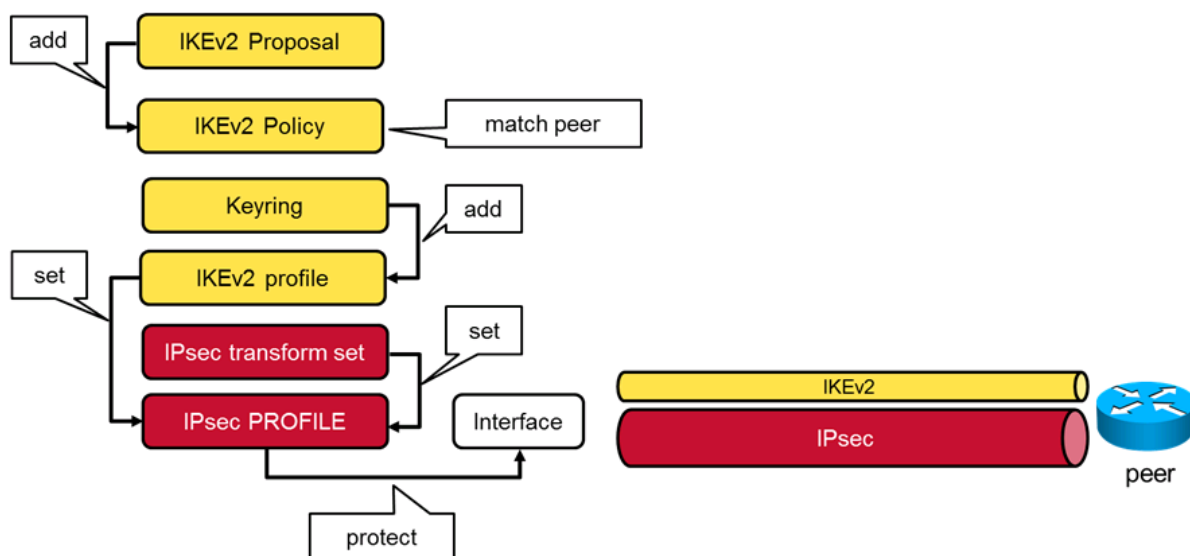
Show Command	Description
R# show crypto ipsec transform-set default	Display default setting of Transform Set
R# show crypto ipsec profile default	Display default setting of IPSec Profile
R# show crypto ikev2 proposal default	Display default IKEv2 Proposal
R# show crypto ikev2 policy default	Display default IKEv2 Policy
R# show crypto ikev2 authorization policy default	Display default IKEv2 Authorization Policy

Modifying defaults	R1(config)#crypto ikev2 proposal default R1(config-ikev2-proposal)# encryption aes-cbc-128 R1(config-ikev2-proposal)# Integrity md5 R1(config)# crypto ipsec transform-set default esp-aes 256 esp-sha-hmac
Restoring defaults	R1(config)# default crypto ikev2 proposal R1(config)# default crypto ipsec transform-set
Disabling defaults	R1(config)# no crypto ikev2 proposal default R1(config)# no crypto ipsec transform-set default

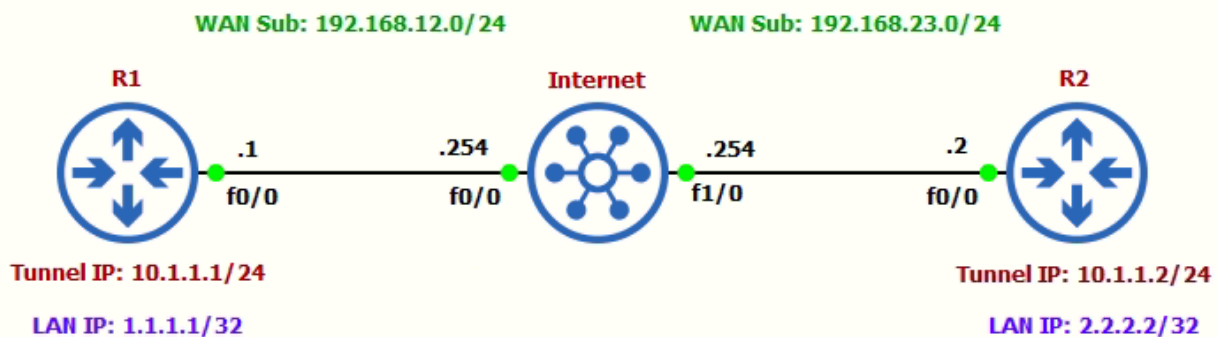
Configuration Components:

The following IKEv2/IPSec configuration components are required for FlexVPN.

Component	Description
IKEv2 Proposal	Mandatory Defines encryption, integrity algorithm, DH group used for protection. Can specify multiple entries for each option. Authentication method and SA lifetime NOT contained in proposal.
IKEv2 Policy	Mandatory Matches peers & associates the IKEv2 Proposal by binding already created IKEv2 Proposal so it is selected for negotiation with the defined peer.
IKEv2 Keyring	Mandatory only if using PSK authentication. Used to define the pre-shared keys. Unlike IKEv1 this can be asymmetric, one key for the local router and another for the remote router.
IKEv2 Profile	Mandatory Defines the local/remote IKEv2 identities (address/identity). Defines the local/remote authentication type. Defines IKEv2 keyring if using PSK authentication.
IPSec Transform Set	Specifies the acceptable security protocols and algorithms for the IPSec SA.
IPSec Profile	References the IPSec Transform Set if NOT default. References the IKEv2 Profile if NOT default. IPSec Profile attached to the Tunnel interface.



Site-to-Site FlexVPN Configuration (S-VTI):



R1 Router Basic Configuration

```
R1(config)#interface f0/0
R1(config-if)#description Public IP to Internet
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown
R1(config)#interface loopback1
R1(config-if)#description LAN IP
R1(config-if)#ip add 1.1.1.1 255.255.255.255
R1(config-if)#no shutdown
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.12.254
```

R2 Router Basic Configuration

```
R2(config)#interface f0/0
R2(config-if)#description Public IP to Internet
R2(config-if)#ip add 192.168.23.2 255.255.255.0
R2(config-if)#no shutdown
R2(config)#interface loopback1
R2(config-if)#description LAN IP
R2(config-if)#ip add 2.2.2.2 255.255.255.255
R2(config-if)#no shutdown
R2(config)#ip route 0.0.0.0 0.0.0.0 192.168.23.254
```

Internet Router Basic Configuration

```
ISP(config)#interface f0/0
ISP(config-if)#description Public IP to R1
ISP(config-if)#ip add 192.168.12.254 255.255.255.0
ISP(config-if)#no shutdown
ISP(config)#interface f1/0
ISP(config-if)#description Public IP to R2
ISP(config-if)#ip add 192.168.23.254 255.255.255.0
ISP(config-if)#no shutdown
```

Router R1 FlexVPN Configuration

```
R1(config)#ip domain-name test.com
R1(config)#crypto ikev2 keyring KR
R1(config-ikev2-keyring)#peer R2
R1(config-ikev2-keyring-peer)#address 192.168.23.2
R1(config-ikev2-keyring-peer)#pre-shared-key local cisco
R1(config-ikev2-keyring-peer)#pre-shared-key remote cisco
R1(config-ikev2-keyring)#crypto ikev2 profile PRO1
R1(config-ikev2-profile)#match identity remote fqdn R2.test.com
R1(config-ikev2-profile)#identity local fqdn R1.test.com
R1(config-ikev2-profile)#authentication local pre-share
R1(config-ikev2-profile)#authentication remote pre-share
R1(config-ikev2-profile)#keyring local KR
R1(config)#crypto ipsec profile default
R1(ipsec-profile)#set ikev2-profile PRO1
R1(config)#interface tunnel0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#tunnel source fastEthernet 0/0
R1(config-if)#tunnel destination 192.168.23.2
R1(config-if)#tunnel protection ipsec profile default
```

Router R2 FlexVPN Configuration

```
R2(config)#ip domain-name test.com
R2(config)#crypto ikev2 keyring KR
R2(config-ikev2-keyring)#peer R1
R2(config-ikev2-keyring-peer)#address 192.168.12.1
R2(config-ikev2-keyring-peer)#pre-shared-key local cisco
R2(config-ikev2-keyring-peer)#pre-shared-key remote cisco
R2(config-ikev2-keyring)#crypto ikev2 profile PRO1
R2(config-ikev2-profile)#match identity remote fqdn R1.test.com
R2(config-ikev2-profile)#identity local fqdn R2.test.com
R2(config-ikev2-profile)#authentication local pre-share
R2(config-ikev2-profile)#authentication remote pre-share
R2(config-ikev2-profile)#keyring local KR
R2(config)#crypto ipsec profile default
R2(ipsec-profile)#set ikev2-profile PRO1
R2(config)#interface tunnel0
R2(config-if)#ip address 10.1.1.2 255.255.255.0
R2(config-if)#tunnel source fastEthernet 0/0
R2(config-if)#tunnel destination 192.168.12.1
R2(config-if)#tunnel protection ipsec profile default
```

Routing on Site-To-Site FlexVPN:

Static Routing

R1(config)#ip route 2.2.2.2 255.255.255.255 tunnel0

R2(config)#ip route 1.1.1.1 255.255.255.255 tunnel0

R1 Dynamic Routing Protocol

R1(config)#router eigrp 1

R1(config-router)#network 1.1.1.1 0.0.0.0

R1(config-router)#network 10.1.1.0 0.0.0.255

R2(config-router)#no auto-summary

R2 Dynamic Routing Protocol

R2(config)#router eigrp 1

R2(config-router)#network 2.2.2.2 0.0.0.0

R2(config-router)#network 10.1.1.0 0.0.0.255

R2(config-router)#no auto-summary

R1#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/fivr	Status
1	192.168.12.1/500	192.168.23.2/500	none/none	READY

Encr: AES-CBC, keysize: 256, Hash: SHA512, DH Grp:5, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/340 sec

R1#show crypto ipsec sa

interface: Tunnel0
Crypto map tag: Tunnel0-head-0, local addr 192.168.12.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.12.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (192.168.23.2/255.255.255.255/47/0)
current_peer 192.168.23.2 port 500
PERMIT, flags={origin is acl,}
#pkts encaps: 25, #pkts encrypt: 25, #pkts digest: 25
#pkts decaps: 25, #pkts decrypt: 25, #pkts verify: 25
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 192.168.12.1, remote crypto endpt.: 192.168.23.2
path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
current outbound spi: 0x1F51BD97(525450647)
PFS (Y/N): N, DH group: none

inbound esp sas:
spi: 0xCA24C565(3391407461)
transform: esp-aes esp-sha-hmac ,
in use settings ={Transport,}
conn id: 4, flow_id: 4, sibling_flags 80000000, crypto map: Tunnel0-head-0
sa timing: remaining key lifetime (k/sec): (4322178/3165)
IV size: 16 bytes
replay detection support: Y
status: ACTIVE(ACTIVE)